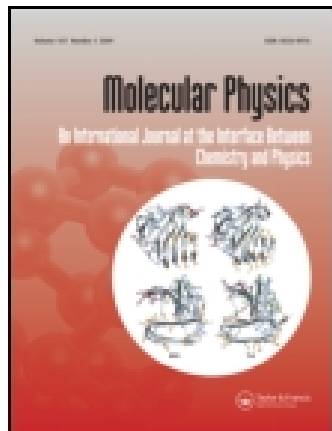




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Spectroscopic and transport properties of water

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Spectroscopic and transport properties of water Model calculations and the interpretation of experimental results

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Computer simulation by the method of molecular dynamics has been used to study the dynamical properties of liquid water at several state points. The interactions between molecules were represented by the potential energy function of Matsuoka, Clementi and Yoshimine. The main effort has been directed at extracting from the simulations results of significance for the interpretation of measurements of spectroscopic and transport properties of real water, including isotope effects. Results are reported for the self diffusion coefficient, N.M.R. relaxation times, infrared and Raman spectra, and functions relevant to coherent and incoherent inelastic neutron scattering and to dielectric relaxation. A basis is provided for the explanation of a number of experimental observations.

1. INTRODUCTION

There is now a substantial history of computer simulations of liquid water based on pairwise additive models of the intermolecular potential [1]. Much of this work has been concerned with the refinement of potential models by comparison with a limited range of properties (largely structural) of real water. Some success has been achieved and a number of models have been developed which satisfactorily reproduce those features of the structure which are generally ascribed to hydrogen bonding between molecules. Given the existence of such models, it should be possible to use computer simulation in a different sense, namely in the interpretation of some of the distinctive properties of water (particularly of its dynamics), the origins of which are not fully understood at the microscopic level. This is one of the main themes of the present paper. As a corollary, such a study should serve to identify those properties of water which are sensitive to details of the interactions between molecules and are hence of particular value in the refinement of intermolecular potentials.

In the pioneering molecular dynamics study of Rahman and Stillinger [2] the potential used was a four point charge model due to Ben Naim and Stillinger (BNS). A suitably scaled version of the potential (SBNS) was found to yield a fair description of many of the properties of real water ; the structure of the

simulated liquid was found to be that of a tetrahedrally ordered, random and highly strained hydrogen bonded network. With the experience gained in this work, an improved four point charge model was developed, the so-called ST2 potential [3]. The latter has subsequently been widely studied and is probably the best known model of the molecular interactions in bulk water. However, it is generally considered that the ST2 potential leads to a local order in the liquid which is overly tetrahedral in comparison with real water. Other potential models which have been studied by molecular dynamics include the central force model of Lemberg and Stillinger [4], a revised version (RCF) of the Lemberg–Stillinger potential [5] and a three charge model due to Berendsen *et al.* (3CM) [6].

Over the same period, a large number of Monte Carlo calculations have also been reported. Indeed, the first published account of an attempt to simulate liquid water was a paper by Barker and Watts [7] describing the results of a Monte Carlo calculation for a potential fitted to gas phase properties by Rowlinson [8]. The ST2 potential has also been studied by Monte Carlo [9, 10] and there has been a recent attempt to extend the scope of the calculations through the explicit incorporation of the polarizability of the molecule [11]. However, the bulk of the published Monte Carlo calculations have been based on the configuration interaction potential of Matsuoka, Clementi and Yoshimine (MCY) [12], derived by fitting to the results of *ab initio* quantum mechanical calculations on the dimer $(\text{H}_2\text{O})_2$. The potential energy surface was parameterized in terms of three atom–atom potentials, supplemented by the electrostatic interaction between charge distributions consisting of charges $-\frac{1}{2}Q$ at the site of each hydrogen atom and a charge Q placed on the main symmetry axis at a distance δ from the oxygen atom and towards the protons. The values chosen for $Q(= -1.435|e|)$, $\delta(=0.2577 \text{ \AA})$ and the bond lengths $d_{\text{OH}}(=0.9572 \text{ \AA})$ and $d_{\text{HH}}(=1.5137 \text{ \AA})$ correspond to a monomer dipole moment of 2.19 D, compared with the measured value in the gas phase of 1.85 D.

It was shown at an early stage [13] that the MCY potential accounts satisfactorily for the distribution of oxygen atoms in water at room temperature. Although there are some differences in detail, the results obtained for the oxygen–oxygen radial distribution function $g_{\text{OO}}(R)$ are in generally good agreement with those derived from X-ray scattering experiments [14]. Recently, Impey *et al.* [15] have shown that the MCY potential also gives good results for the changes which occur in $g_{\text{OO}}(R)$ as the temperature is increased, particularly with respect to the disappearance of the peak due to second neighbours. Satisfactory agreement was also found with the results of neutron scattering experiments on D_2O [16], which are sensitive primarily to the correlations described by the radial distribution functions $g_{\text{OD}}(R)$ and $g_{\text{DD}}(R)$. The agreement with the neutron results was found to include the changes in the scattering pattern which appear when the liquid is subjected to a high external pressure [17]; these changes reflect the distortion of the hydrogen bond network which occurs under compression. Finally, the overall level of agreement with the X-ray experiments was found to be significantly improved if the pair potential energy was uniformly scaled by a factor of 1.14, equivalent to a scaling of temperature (in the simulation) by approximately 0.88.

Given the generally good results obtained for the structure of the liquid, it is surprising that so little attention has been paid to the dynamical properties of the

MCY model; the only work of which we are aware is that contained in a brief note by Rapaport and Scheraga [18]. The purpose of the present paper is to report the results of a series of molecular dynamics calculations which enable us to assess the extent to which the MCY potential can account for the available experimental data on the motion of water molecules. As we have already hinted, the results of such calculations should also be of help in improving our understanding of the properties of real water. A longer term motivation for the work is our interest in using the MCY potential in the simulation of aqueous solutions. Clementi and co-workers [19] have calculated ion-water potentials by *ab initio* methods for a number of simple ions. For the study of ionic solutions it is natural to combine these potentials with the MCY model of the water-water interactions, but for results to be useful it is clearly necessary for the model solvent to be a realistic one.

2. MOLECULAR DYNAMICS CALCULATIONS

For the most part, the molecular dynamics calculations we have undertaken have followed conventional lines. All the results reported here were obtained for systems of 125 molecules with the usual cubic periodic boundary conditions. Only in two respects have we departed in an important way from the methods used in most previous calculations on liquid water. First, the electrostatic forces and energies were calculated by the Ewald method. The question of how best to handle the long range interactions characteristic of polar systems is one which has been keenly debated and the relative merits of a number of different schemes (spherical cut-off, minimum image, reaction field and Ewald) have been discussed [20]. The problem has not been fully resolved, but there is now a body of evidence which suggests that use of the Ewald method, though not without its disadvantages, does better justice than other methods to the dipolar correlations (and hence to the dielectric properties) expected in the infinite system limit [21]. The Ewald method also has a minor but not insignificant advantage over the more commonly used spherical cut-off scheme in that it avoids the energy drift associated with the passage of molecules through the surface of the truncation sphere.

The second technical detail that should be mentioned is that we have worked in general with a timestep of 0.0015 ps. This is more than three times larger than that which can conveniently be used when the equations of motion are solved in terms of Euler angles, as was the case in the original work of Rahman and Stillinger [2]. The improvement has been achieved by use of the method of constraints devised by Ryckaert *et al.* [22]. A comparable improvement over earlier work has been achieved by Rapaport and Scheraga [18], who used a method of integration of the equations of motion due to Evans and Murad [23]. The particular choice of timestep in the present work was dictated by our interest in following the reorientational motion of the molecules. An acceptable level of energy conservation, which is the usual monitor of the accuracy of a molecular dynamics calculation, is obtained even with a timestep as large as 0.003 ps.

Details of the state conditions in the seven molecular dynamics runs we shall discuss (five for H₂O and two for D₂O) are given in table 1. Also included in the table are the calculated mean potential energy U , the compressibility factor

Table 1. Details of the molecular dynamics calculations.

Run	T/K	$V/\text{cm}^3 \text{ mol}^{-1}$	$-U/\text{kJ mol}^{-1}$	$pV/Nk_B T$	$D/10^{-9} \text{ m}^2 \text{ s}^{-1}$
1	242	18.00	38.6	9.1	0.50 ± 0.08
2	282	18.00	36.5	8.1	1.81 ± 0.07
3†	284	18.10	36.4	8.1	1.26 ± 0.14
4	286	18.00	36.1		2.12 ± 0.15
5	294	18.00	35.9	8.5	2.3 ± 0.2
6†	304	18.13	35.4		2.28 ± 0.10
7	360	18.00	32.7	7.7	5.1 ± 0.2

† D_2O . Other calculations are for H_2O .

$pV/Nk_B T$, and the self diffusion coefficient D . Each of the runs was of approximately 6000 timesteps (about 10 ps).

3. STATIC PROPERTIES

Several groups of workers have reported results for U for the MCY potential and it is well established that the simulated liquid is considerably too weakly bound. Indeed, the scaling of the potential used in [15] was suggested by this fact. The present work serves only to confirm the earlier findings; for comparison, the experimental value at 25°C is $-41.8 \text{ kJ mol}^{-1}$. However, the same weakness of the model is evident also in the results given here for the equation of state, of which no systematic study has previously been reported. In the original Monte Carlo work of Lie *et al.* [13], the pressure (at 25°C) was said to be 'greater than 3000 atm'. The origin of this estimate is not explained, but in any event the true situation is even less satisfactory than the statement suggests. In the NpT-ensemble Monte Carlo calculations of Owicki and Scheraga [24], the calculated volume at 1 atm and 25°C was $23.8 \text{ cm}^3 \text{ mol}^{-1}$, which is 32 per cent larger than the experimental value. Given the estimates made of the compressibility of MCY water (38 to $53 \times 10^{-6} \text{ atm}^{-1}$) in the relevant range of density [24], this result is clearly inconsistent with that of Lie *et al.* [13], particularly because Owicki and Scheraga [24] include a correction for long range dispersion interactions which has the effect of reducing the calculated volume. Similarly, Morse and Rice [25] have recently shown that although the MCY potential yields the correct structure of several forms of ice, the predicted densities are all about 20 per cent too low, even when the dilation due to zero point energy is ignored. The values of $pV/Nk_B T$ given in table 1 are in generally good accord with the results of [24] and [25], since they correspond to pressures ranging from 10.0 (Run 1) to 12.6 (Run 7) kbar. Overall, therefore, despite some encouraging results obtained for the compressibility [24] and heat capacity [15], the MCY potential gives a poor description of the thermodynamic properties of water.

Another quantity of considerable interest is the static dielectric constant ϵ . DeLeeuw *et al.* [26] have shown that for simulations based on the Ewald method, ϵ is related to the mean square moment $\mathbf{M}$ of the primitive cube by the formula

$$\epsilon = 1 + 12\pi \langle \mathbf{M} \cdot \mathbf{M} \rangle / V k_B T \quad (1)$$

(remembering that $\langle \mathbf{M} \rangle = 0$). However, equation (1) yields quantitatively reliable results only when the Ewald sums are calculated with a high accuracy [20]. There is, in addition, the usual difficulty associated with the reliable estimation of variances from small samples. Thus the calculation of ϵ from equation (1) is subject to considerable uncertainty. The simulations summarized in table 1 were largely carried out with the real space term in the Ewald sum truncated at one half the length of the primitive cube and the Fourier series truncated after the first 353 vectors. Though adequate for a qualitative study of dipolar correlations, this truncation scheme was not considered to correspond to an accuracy sufficiently high to warrant a calculation of the dielectric constant. In Runs 5 and 6, on the other hand, the Fourier series was truncated after 754 vectors, which extends to beyond the main peak in the oxygen–oxygen structure factor. The mean dielectric constant over the two runs was $\epsilon = 50 \pm 10$, compared with the experimental value of $\epsilon \simeq 80$. We shall return to a discussion of the dielectric properties of the model in § 8.

We have also computed the three partial (atom–atom) radial distribution functions which characterize the structure of water. These functions have been extensively discussed elsewhere [13, 15], particularly in relation to X-ray and neutron scattering experiments. We therefore omit any further consideration of them here.

4. SELF DIFFUSION COEFFICIENTS

The calculated self-diffusion coefficients reported in table 1 were obtained by monitoring the displacement of a labelled atom i as a function of time, averaging over all atoms and all choices of time origin, and exploiting the Einstein relation

$$6D = \lim_{t \rightarrow \infty} \frac{d}{dt} \langle |\mathbf{R}_i(t) - \mathbf{R}_i(0)|^2 \rangle. \quad (2)$$

We have applied this relation separately to the trajectories of both the hydrogen (or deuterium) atoms and the oxygen atoms. This provides a useful check on the reliability of the calculated diffusion coefficients, since in the long time limit the derivative must be the same for both atomic species.

In figure 1 we compare our results for H_2O with the experimental data of Krynicki *et al.* [27] (obtained by the proton spin echo method) and with previous molecular dynamics calculations for a variety of potential models. It is clear from the figure that the results of the present study are in good agreement with those obtained for the MCY model by Rapaport and Scheraga [18], particularly when account is taken of differences in density between the two calculations. Moreover, the MCY results agree well with experiment in the region around room temperature, but the temperature dependence is poorly reproduced. In particular, the calculated values at high temperatures are significantly too low. The same trend is evident, though less marked, in the results for the ST2 potential. Indeed, to date, only the scaled BNS potential (the precursor of the ST2 potential) seems able to yield both the correct absolute values and correct temperature dependence of the diffusion coefficient.

Two other points concerning the MCY results are worthy of note. First, the fact that the calculated plot of D versus T crosses the experimental curve means that there is no possibility of successfully scaling the results in the manner

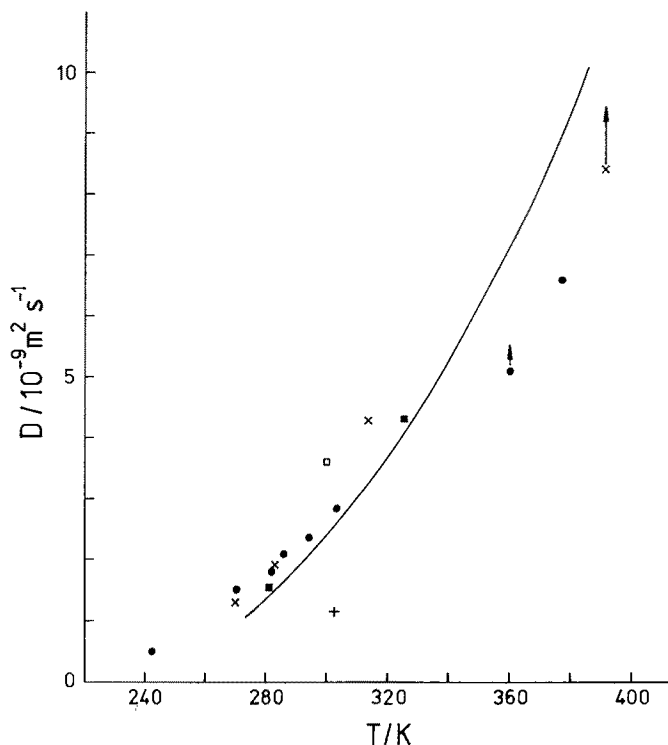


Figure 1. Diffusion coefficient of water. The curve shows the experimental results along the s.v.p. line [27] and the points show the results of molecular dynamics calculations for the MCY (filled circles), ST2 (x), SBNS (filled squares), RCF (+) and 3CM (open square) potential models. The MCY results for 271, 304 and 377 K are taken from [18] and those for the other temperatures are from the present work. The arrows indicate roughly the corrections required to convert from the density used in the molecular dynamics calculations to that appropriate to s.v.p. conditions.

found appropriate for structural properties [15]. Secondly, the discrepancy with experiment at high temperatures suggests that the weakening of the hydrogen bond network with increasing temperature is inadequately reproduced by the model.

From table 1 and [18] we obtain for the ratio of the self diffusion coefficients in normal and heavy water the values $D(\text{H}_2\text{O})/D(\text{D}_2\text{O}) = 1.56 \pm 0.25$ (at 286 K) and 1.20 ± 0.15 (at 303 K). If the only effect of isotopic substitution were to change the molecular mass M , the diffusion coefficient would be expected to scale as $M^{-1/2}$. This would yield a self diffusion coefficient ratio that was independent of temperature and given by $D(\text{H}_2\text{O})/D(\text{D}_2\text{O}) = 1.054$. In the case of water, of course, a more important effect of deuteration is the change in the moments of inertia of the molecule. No simple dependence on M can therefore be expected, since the breaking of hydrogen bonds which accompanies the process of diffusion must result in a large translation-rotation coupling. This effect is one which can be understood in classical terms. There is, in addition, a possible mechanism of quantum mechanical origin, namely the differing extent to which it is necessary to take account of zero point motion in liquid H_2O and D_2O .

However, our results for the ratio of the two diffusion coefficients, though subject to large uncertainties, are consistent with the experimental values reported by Mills [28], who finds $D(\text{H}_2\text{O})/D(\text{D}_2\text{O}) = 1.294$ at 278 K, 1.228 at 298 K and 1.200 at 318 K. The comparison is certainly not conclusive, but it does suggest that the observed departures from the $M^{-1/2}$ rule in real water are primarily a classical inertial effect.

As an alternative to the use of equation (2), the self diffusion coefficient may be calculated from the time integral of either the proton or oxygen velocity autocorrelation function. These functions are shown in figure 2 for the case of H_2O at 286 K (Run 4 of table 1) and are similar to those obtained in other simulations of liquid water. Not unexpectedly, the proton function $Z_{\text{H}}(t)$ is the more oscillatory at short times and the oxygen function $Z_{\text{O}}(t)$ is the longer lived.

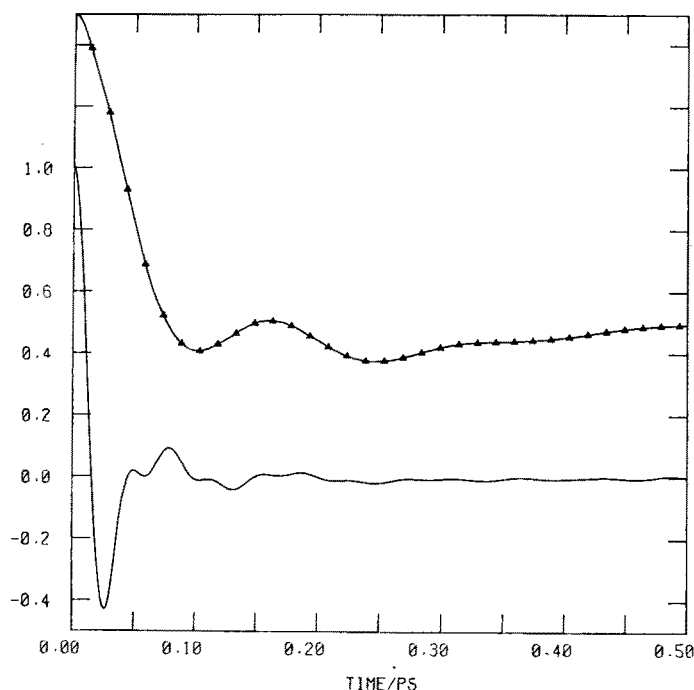


Figure 2. Atomic velocity autocorrelation functions for H_2O at 286 K. Top : oxygen atoms ; bottom : hydrogen atoms.

To obtain further insight into the microscopic diffusive process, we have also calculated the time autocorrelation functions for the components of the centre of mass velocity in the coordinate system defined by the molecular orientation at the initial time, that is

$$Z^\alpha(t) = \langle v_i^\alpha(t) v_i^\alpha(0) \rangle \quad (3)$$

where i labels a molecule and the velocity component v_i^α is defined as

$$v_i^\alpha(t) = \mathbf{e}_i^\alpha(0) \cdot \mathbf{v}_i(t) \quad \alpha = x, y, z \quad (4)$$

where $\mathbf{v}_i(t)$ is the molecular velocity at time t . The unit vectors $\mathbf{e}^x$, $\mathbf{e}^y$ and $\mathbf{e}^z$ are defined by figure 3.

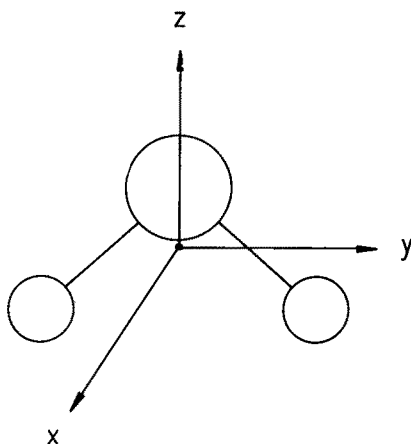


Figure 3. Definition of the coordinate system used in the discussion of molecular motion.

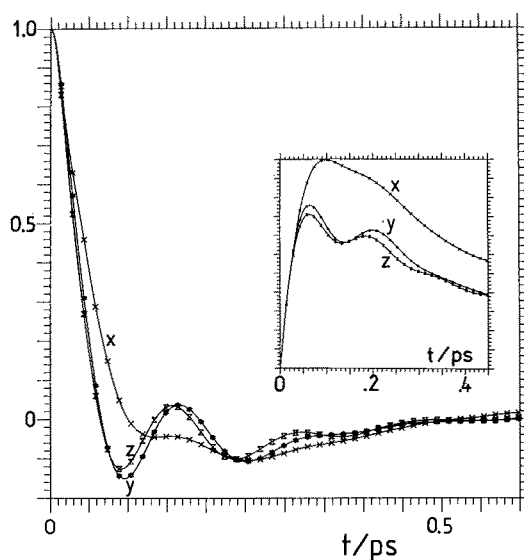


Figure 4. Autocorrelation functions of the components of the centre of mass velocity with respect to the molecular axes at the initial time. The inset shows the running integrals of the three functions (arbitrary units). The results are for H_2O at 286 K.

The three functions $Z^x(t)$, $Z^y(t)$ and $Z^z(t)$ for H_2O at 286 K are plotted in figure 4; it should be noted that, over the timescale of interest, molecular reorientation does not play an important rôle. Two distinct oscillation frequencies are discernible. The higher frequency is the more prominent in $Z^y(t)$ and $Z^z(t)$; it is also weakly visible in $Z^x(t)$, but in this case the dominant effect is the slower fluctuation. The two frequencies may be correlated with spectroscopically observable features, as we shall see in later sections. The two functions $Z^y(t)$

and $Z^z(t)$ resemble each other closely, but $Z^x(t)$ is very different. The three running integrals are shown in the inset to the figure. From these curves it is clear that on the timescale under consideration the molecules displace preferentially in the x -direction, that is in a direction perpendicular to the plane of the molecule.

5. DYNAMICS OF REORIENTATION

5.1. General discussion

To discuss the reorientational motion of the molecules, we use again the coordinate system specified in figure 3. In figure 5 we show the results obtained for H_2O at 286 K for the time autocorrelation functions $C_l^z(t)$ of the first four Legendre polynomials ($l=1, 2, 3, 4$) of the angle subtended by the molecular z -axis at time t with respect to its position at $t=0$, that is

$$C_l^z(t) = \langle P_l[\mathbf{e}^z(t) \cdot \mathbf{e}^z(0)] \rangle. \quad (5)$$

The functions are plotted in the form $[l(l+1)]^{-1} \ln C_l^z(t)$, since this type of presentation has been found useful in evaluating the merits of models of molecular reorientation [29].

All four functions plotted in figure 5 have the same basic features, the glitch at times so short that they would correspond to almost free inertial motion in simple molecular fluids and the rapid onset of the diffusive, slowly decaying, exponential régime. The results for $l=1$ and $l=2$ are very similar to those

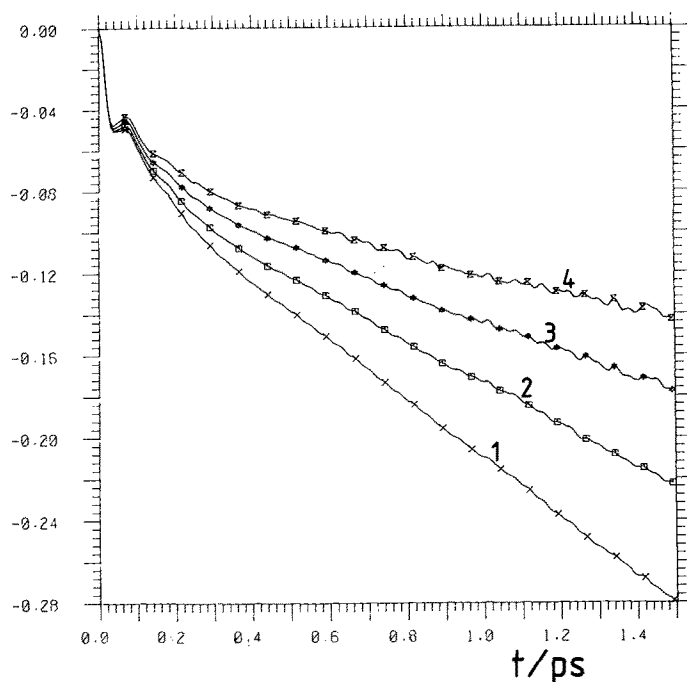


Figure 5. The functions $[l(l+1)]^{-1} \ln C_l^z(t)$ for H_2O at 286 K and the values of l indicated in the figure.

obtained for the BNS potential [2]. Although we describe the motion at long times as diffusive, it is clear that the behaviour there is not Debye-like since the functions, as plotted, do not coincide. Note also that the form in which the results are shown disguises the fact that the amplitude of the glitch increases rapidly with increasing l .

In most dynamical models of liquid water, each molecule is supposed to sit in a long-lived cage of four tetrahedrally coordinated neighbours. The glitch in the orientational correlation functions reflects the librational motion of the central molecule in the cage and can therefore be linked to the prominent librational features which appear in both the infrared and Raman spectra [30]. By contrast, the exponential régime is associated with the structural breakup of the cage. The rate of decay largely determines the N.M.R. relaxation time [31] and its smooth character accounts for the simplicity of the dielectric dispersion in the microwave region [32].

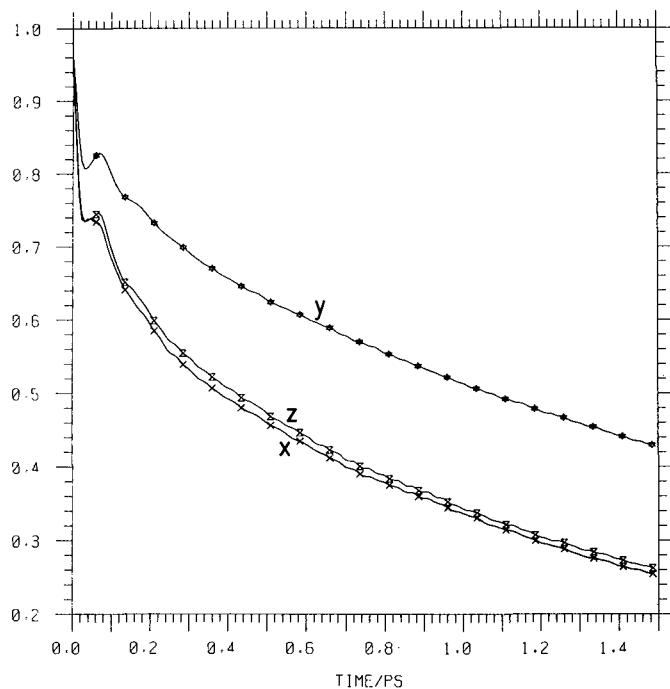


Figure 6. The functions $C_2^x(t)$, $C_2^y(t)$ and $C_2^z(t)$ for H_2O at 286 K.

In figure 6 we plot the $l=2$ autocorrelation functions (again for H_2O at 286 K) for the molecule-fixed x , y and z directions. All three functions have the same general shape and, remarkably, the x and z functions are almost identical; the position of the glitch is much the same in every case. An estimate of the mean angle of libration about the axis α may be obtained by setting

$$P_l(\cos \theta_l^\alpha) = C_l^\alpha(t_g), \quad (6)$$

where t_g is the time corresponding to the minimum in the glitch. The results turn out to be almost independent of l , as is to be expected from figure 5. The

spread in librational angle is therefore very small. For this particular state point we find that $\theta_i^x = 25^\circ$, $\theta_i^y = 21^\circ$ and $\theta_i^z = 25^\circ$. Libration about x and z is therefore greater than that about the y -axis.

For the purposes of discussion, it is useful to relate the behaviour of the orientational correlation functions to that of the angular velocity correlation functions. In the present case we find that a first order cumulant expression [33] provides a good fit to the data at short times; the higher order cumulants account for the fanning out at long times of the curves plotted in figure 5. At the simplest level of approximation, the first order autocorrelation functions $C_1^\alpha(t)$ are given by

$$C_1^\alpha(t) = \exp \left\{ - \int_0^t (t-\tau) [\langle \omega^\beta(t) \omega^\beta(0) \rangle + \langle \omega^\gamma(t) \omega^\gamma(0) \rangle] d\tau \right\}, \quad (7)$$

where $\alpha \neq \beta \neq \gamma$. The more complex expressions for the higher order functions may be found elsewhere [33]; however, the arguments given below for $l=1$ apply also to higher values of l . The angular velocity components ω^β , ω^γ which appear in equation (7) are referred to the coordinate system which coincides with the molecular frame at $t=0$. In practice, at low temperatures the angular

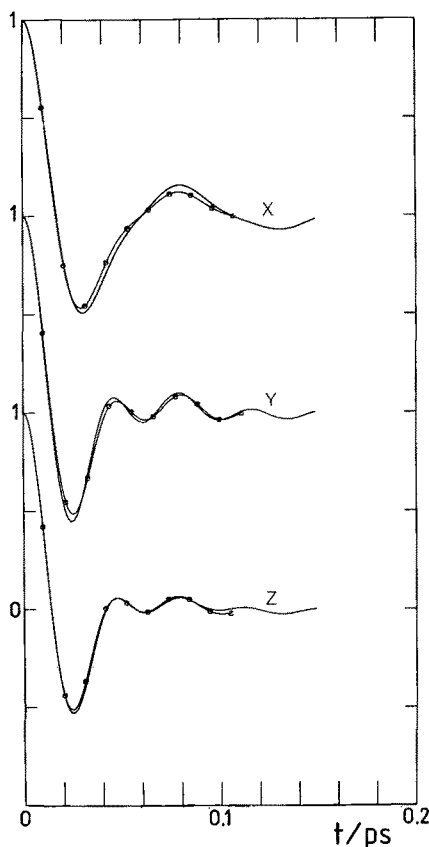


Figure 7. Autocorrelation functions of the components of the angular velocity with respect to the molecular axes at the initial time. The curves without points are the results for H₂O at 286 K and the curves with points are the scaled functions for D₂O (see text).

velocity autocorrelation functions defined in this way are almost identical to those referred to the instantaneous body-fixed frame (the more usual choice of coordinate system). This implies that the angular velocity autocorrelation functions are torque driven and not strongly influenced by inertial effects, that is that the precessional terms in the Euler equations play only a minor rôle even at short times. The three angular velocity autocorrelation functions obtained for H₂O at 286 K are shown in figure 7; the thermodynamic state and isotope dependence will be discussed below.

The form of equation (7) shows explicitly that the glitch in the orientational correlation functions is linked to the large negative region in the angular velocity functions. For a certain range of t , the integral in (7) becomes negative. Thus, after a rapid initial decay, the orientational correlation function must grow as t increases.

In view of the fact that $\langle \omega^x(t)\omega^x(0) \rangle$ differs so markedly from $\langle \omega^y(t)\omega^y(0) \rangle$ and $\langle \omega^z(t)\omega^z(0) \rangle$, as shown in figure 7, it is at first sight puzzling that the orientational autocorrelation functions $C_l^\alpha(t)$ for different α should be so similar. The reason that the distinctive character of the ω^x function does not manifest itself is connected with the properties of the inertia tensor. For the H₂O molecule, $I_{xx} \simeq \frac{3}{2}I_{zz} \simeq 3I_{yy}$. Since the equal time values of the angular velocity autocorrelation functions are given by

$$\langle \omega^\alpha \omega^\alpha \rangle = k_B T / I_{\alpha\alpha} \quad (8)$$

it follows that $\langle \omega^x(t)\omega^x(0) \rangle$ is swamped by $\langle \omega^y(t)\omega^y(0) \rangle$ in $C_l^z(t)$ (see equation (7)) and contributes only weakly to $C_l^y(t)$; $C_l^x(t)$ and $C_l^z(t)$ should therefore be very similar, as is seen to be the case for $l=2$ in figure 6. It should be noted that, in the case of the BNS potential [2], it is the function $\langle \omega^y(t)\omega^y(0) \rangle$ which differs markedly from the other two. In terms of the functions $C_l^\alpha(t)$, the reorientational motion should therefore appear to be less isotropic than it does for the MCY model and thus in better agreement with experiment (see below).

5.2. Comparison with experiment

The available experimental information on the low frequency reorientational motion of water molecules is not extensive. Nothing can be learnt from a study of the spectroscopic lineshapes corresponding to internal vibrations, since these are dominated by vibrational relaxation. The reorientational component of the depolarized Rayleigh spectrum is weak, because the polarizability tensor is nearly isotropic, and it merges with a collision induced component. The dielectric relaxation in the microwave region is strongly collective in character and not easily related to the reorientation of single molecules. The intramolecular part of the dipole-dipole relaxation of the proton N.M.R. signal yields only the zero frequency Fourier component of the transform of $C_2^y(t)$. Further information can in principle be obtained from studies of the quadrupole relaxation of the deuterium nucleus, but the quadrupole coupling tensor is poorly characterized in the liquid.

Orientational correlation functions such as those pictured in figure 5 may conveniently be Fourier transformed by first writing them in the form

$$C_l^\alpha(t) = \phi_l^\alpha(t) + A_l^\alpha \exp(-t/\tau_l^\alpha). \quad (9)$$

In the present work, the parameters A_l^α and τ_l^α have been determined by

Table 2. Reorientational correlation times (in ps).

Run†	τ_1^x	τ_1^y	τ_1^z	τ_2^x	τ_2^y	τ_2^z	$\tau_{N.M.R.}$
1	9.0	16.3	10.3	4.3	7.0 (5.7)	4.4	
2	3.2	6.0	3.5	1.8	2.7 (2.1)	1.7	3.6‡
3	3.6	6.3	3.6	2.0	2.8 (2.2)	1.8	4.8§
4	3.1	5.7	3.8	1.7	2.7 (2.0)	1.8	
7	1.0	1.9	1.2	0.6	0.9 (0.6)	0.6	0.9

† For details of the state conditions, see table 1. Bracketed numbers are results for $A_2^y \tau_2^y$. Experimental results for $\tau_{N.M.R.}$ are from [31]: ‡ H₂O at 283 K; § D₂O at 283 K; || H₂O at 363 K.

fitting $C_l^\alpha(t)$ to an exponential at large t (0.8 to 1.5 ps); $\phi_l^\alpha(t)$ is a rapidly decaying, oscillatory function which can be transformed numerically and the long time part can be treated analytically.

Because $\phi_2^y(t)$ is a function which decays rapidly, the value of the transform of $C_2^y(t)$ at $t=0$ is given to a good approximation by $A_2^y \tau_2^y$. For H₂O this quantity should correspond to the relaxation time $\tau_{N.M.R.}$, relaxation time which is associated with intramolecular dipole-dipole coupling. A comparison with experimental data [31] is made in table 2. The agreement is only fair, the calculated values being systematically too low. In addition, the temperature dependence is not sufficiently strong. In the context of a Stokes-Einstein description of the molecular motion, that is one in which both D and $1/\tau_2^\alpha$ are proportional to the shear viscosity [34], this behaviour is clearly consistent with that noted previously for the self diffusion coefficient (§ 4).

The reorientation times τ_1^α and τ_2^α given in table 2 show a weak isotope effect for $\alpha=x$ and y ; similar behaviour is found for $l=3$ and 4. In the case of τ_2^α , the extent to which the reorientation time lengthens on deuteration is in broad agreement with the N.M.R. results of Jonas [31]. However, the strong state dependence of the deuterium coupling constant for water, and the difficulty in separating the inter and intramolecular contributions to the relaxation, preclude any detailed numerical comparison. Hasted [32] has given values for $\tau_1^z(D_2O)/\tau_1^z(H_2O)$ inferred from dielectric relaxation measurements. At 10°C, the quoted ratio is 1.3, which is significantly larger than that suggested by the results of table 2. The isotope effect on reorientation has been linked by Hasted [32], via a Stokes-Einstein relation, to an increase in viscosity on deuteration. Since our calculated translational diffusion coefficients do show a large isotope effect, we suspect that the disagreement in the case of the reorientation time arises from systematic errors associated with the fact that we have fitted our results only to data relating to intermediate times.

Although the low frequency reorientational behaviour is less well characterized than it is for many non-associated liquids, a great deal is known [30] from the infrared and Raman bands between 300 and 900 cm⁻¹ about the higher frequency, librational motion of the water molecules. We shall suppose, for simplicity, that these spectra can be adequately treated as being purely reorientational in origin, neglecting interaction induced moments and assuming that the weight given to motion about different axes is determined by the gas phase dipole

moment and polarizability tensor. Furthermore, although the spectra are collective in origin, we shall suppose that at the high frequencies of interest the dominant contribution comes from the single molecule orientational correlation functions ; we shall see in § 8 that this is indeed true in the infrared case. With these approximations, the infrared spectrum is given by

$$I^{\text{ir}}(\omega) \propto (\hbar\omega/2k_{\text{B}}T) \coth(\hbar\omega/2k_{\text{B}}T) \text{Re} \int_0^\infty \exp(i\omega t) C_1^z(t) dt \quad (10)$$

and the depolarized Rayleigh (‘ Raman ’) spectrum by

$$I^{\text{Ram}}(\omega) \propto \text{Re} \int_0^\infty \exp(i\omega t) \langle \sum_a \delta\alpha_{aa} e_x^a(t) e_z^a(t) \sum_b \delta\alpha_{bb} e_x^b(0) e_z^b(0) \rangle dt \quad (11)$$

where $\delta\alpha_{aa} = \alpha_{aa} - \bar{\alpha}$ is the aa -component of the anisotropy in the polarizability tensor related to the molecular axes and e_x^a is the x -component of $\mathbf{e}^a$ referred to the laboratory coordinate system. Note that the cross terms ($a \neq b$) in equation (11) make only a small contribution. Consequently, the correlation function appearing in the integrand is a weighted sum of the functions $C_2^a(t)$.

The calculated spectra for several state points are shown in figures 8 and 9. The peaks which appear at frequencies greater than about 300 cm^{-1} arise from the same physical processes which lead to the glitch in the orientational correlation functions. The spectral features in this region are in several respects in good agreement with experiment. First, the peak positions are almost independent of temperature. Experimentally, this observation has led to the suggestion that

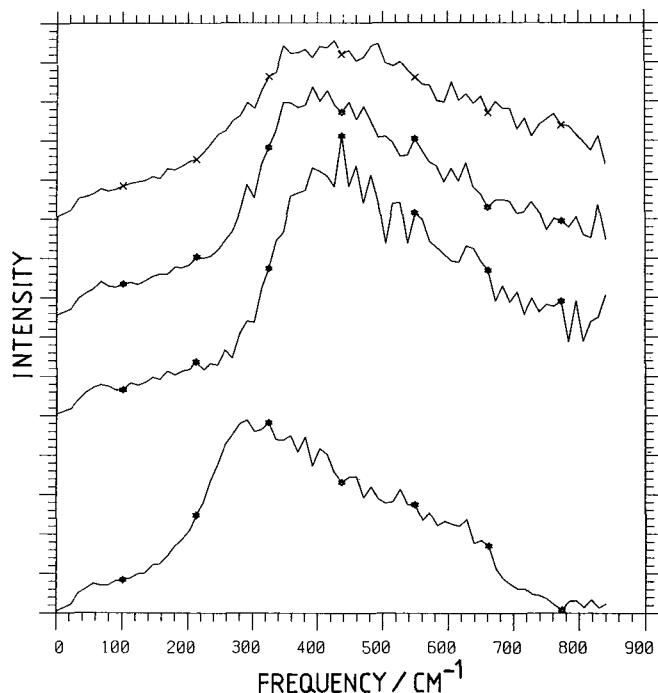


Figure 8. Infrared spectra (arbitrary units). From top to bottom : H_2O at 360 K, H_2O at 286 K, H_2O at 242 K and D_2O at 284 K.

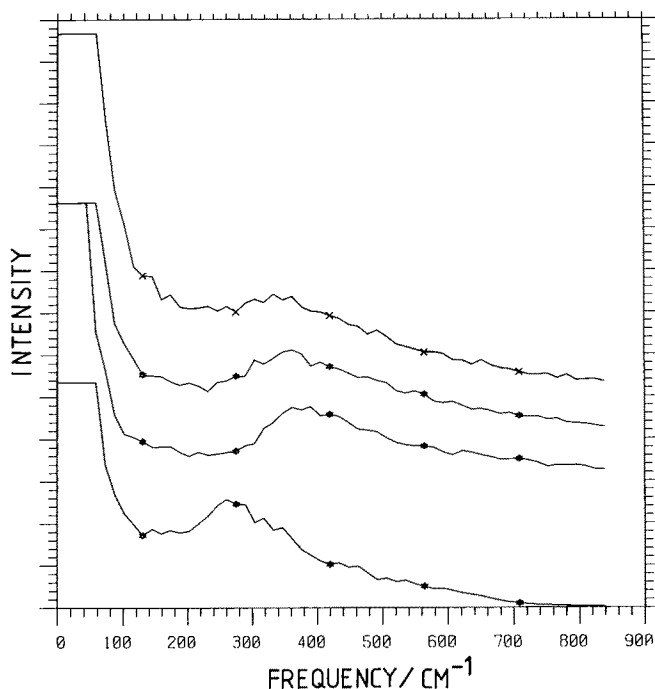


Figure 9. Raman spectra (arbitrary units). From top to bottom : H_2O at 360 K, H_2O at 286 K, H_2O at 242 K and D_2O at 284 K.

liquid water contains clusters of well defined structure [30]. Our results cannot be used to support such an interpretation, since the peaks may equally well be associated with short lived features in the orientational correlation functions. Secondly, the decrease in intensity of the bands as the temperature rises is in semi-quantitative agreement with experiment [30]. This behaviour reflects the fact that the depth of the glitch decreases as the temperature rises because the cage surrounding a given molecule relaxes more rapidly at high temperatures. Finally, the shift in band position on deuteration is also in good agreement with the experimental results [30]. These findings suggest that the MCY model is capable of explaining many of the distinctive librational properties of water molecules which are ascribable to hydrogen bonding. On the other hand, the infrared peak occurs at much too low a frequency in comparison with experiment (at 440 rather than 700 cm^{-1}) and the experimental Raman spectrum shows two peaks (at around 400 and 700 cm^{-1} [30]) rather than the one we have observed.

Some caution is needed in assessing the significance of our results, since the molecular dynamics calculations treat the molecules as being both rigid and classical. Because of quantum dispersion, a real molecule explores the librational potential over a wider range than in the simulations and is therefore more strongly influenced by anharmonicity in the potential. The expected result is that the librational frequencies will be reduced and the effect will be enhanced by the flexibility of the molecules, since the intramolecular geometry will adjust as the molecules librate. Nonetheless, the results do point to certain inadequacies in the MCY model. In particular, the fact that the predicted infrared frequency

is much too low means, according to the discussion following equation (7), that the negative region in $\langle \omega^y(t)\omega^y(0) \rangle$ occurs too late. This in turn suggests that the torques acting about the y -axis are too weak. Moreover, the fact that the experimental Raman spectrum shows two pronounced high frequency peaks implies that in the model the behaviour of $\langle \omega^y(t)\omega^y(0) \rangle$ resembles that of $\langle \omega^z(t)\omega^z(0) \rangle$ too closely; the fault cannot be ascribed to $\langle \omega^x(t)\omega^x(0) \rangle$, since this makes only a small contribution to the functions $C_i^\alpha(t)$.

The experimental infrared and Raman spectra [30] show further peaks at around 60 and 180 cm^{-1} . These occur only weakly, at best, in the calculated spectra of figures 8 and 9. They are features associated with centre of mass motions of the molecules (see the discussion of neutron scattering results below) and do not appear in the calculated spectra because we have not included an intermolecular (interaction induced) component in the polarizability [33].

5.3. Angular velocity autocorrelation functions

As we have already pointed out, the angular velocity autocorrelation functions for the MCY model differ markedly from those for the BNS potentials [2]. In the present case, the y and z functions are similar and x relaxes more *slowly*, whereas in the earlier work the x and z components behaved similarly and the y function decayed more *rapidly*. It is only the z component which has approximately the same form for both models, the Fourier transforms peaking at a little over 400 cm^{-1} . From our earlier arguments, it follows that we must expect considerable differences in the corresponding infrared and Raman spectra.

The structure of the angular velocity autocorrelation functions may be represented schematically as the product of a simple oscillation, $\cos \omega_L t$, and a damping function. The oscillation frequency ω_L is found to be almost independent of temperature; this is a property of real hydrogen bonded liquids, as evidenced [30] by the insensitivity to temperature of the frequencies of peaks in the infrared and Raman spectra. However, the relaxation time τ_R characterizing the damping function decreases rapidly with increasing temperature. This behaviour is correlated with the loss in intensity of the librational peaks as the temperature rises.

On isotopic substitution (D for H), the oscillation frequency shows a marked decrease but the damping time is almost unaltered. As we have already pointed out, the precessional terms in the equations of motion for the angular velocity are small. Thus, to a good approximation, we may write that

$$\dot{\omega}^\alpha = N^\alpha / I_{\alpha\alpha}, \quad (12)$$

where N^α is the torque acting about the molecular axis α . If we now make a short time expansion of the angular velocity autocorrelation function, we find that

$$\langle \omega^\alpha(t)\omega^\alpha(0) \rangle / \langle \omega^\alpha \omega^\alpha \rangle = 1 - t^2 \langle N^\alpha N^\alpha \rangle / 2I_{\alpha\alpha} k_B T + O(t^4). \quad (13)$$

Since isotopic substitution leads only to a small displacement of the centre of mass, the change in mean square torque should also be small. If the short time expansion is valid up to times at which appreciable relaxation of the torque has occurred, the libration frequency of a molecule in a fixed cage of neighbours will scale as the square root of the associated element of the inertia tensor ($I_{xx}(\text{D}_2\text{O}) = 1.931 I_{xx}(\text{H}_2\text{O})$, $I_{yy}(\text{D}_2\text{O}) = 1.80 I_{yy}(\text{H}_2\text{O})$, $I_{zz}(\text{D}_2\text{O}) = 2.00 I_{zz}(\text{H}_2\text{O})$). In fact,

scaling of the time axis of the D_2O functions from Run 3 (284 K) produces a set of curves which are almost identical with those for H_2O at 286 K, as shown in figure 7. What is suprising in this is the fact that not only the frequency ω_L but also the inverse relaxation time τ_R^{-1} scale approximately as $I^{1/2}$. This suggests that the breakup of the coordination cage surrounding a given molecule occurs as the result of molecular reorientation rather than translation. Since the breakup process is the one which determines the relaxation time of the forces and torques, this observation provides a natural explanation of the isotope effects on the transport properties of water.

6. INELASTIC NEUTRON SCATTERING

The high frequency motions seen as peaks in the infrared and Raman spectra may also be observed by inelastic neutron scattering [35]. Information from several experiments, performed under different conditions, may be used to derive the spectral density function, which is the Fourier transform of the velocity autocorrelation function of the nucleus responsible for the scattering; in the case of water, this is the proton. The experimental spectral density function for the protons, $\hat{Z}_H(\omega)$, is shown in figure 11 of [35]; this has well defined peaks at 100, 180 and 570 cm^{-1} . Several groups have, in addition, reported the occurrence of a number of other low frequency peaks [35]. The calculated $\hat{Z}_H(\omega)$ for H_2O at 286 K is plotted in figure 10; this has peaks at about 60 and 190 cm^{-1} , but the dominant feature is the broad band centred at 440 cm^{-1} . There is also a hint of a peak at about 130 cm^{-1} .

The velocity of a given proton may be decomposed into a centre of mass translation and a rotation about the centre of mass in the form

$$\mathbf{v}_H = \mathbf{v}_{CM} + \boldsymbol{\omega} \times \mathbf{R}_H, \quad (14)$$

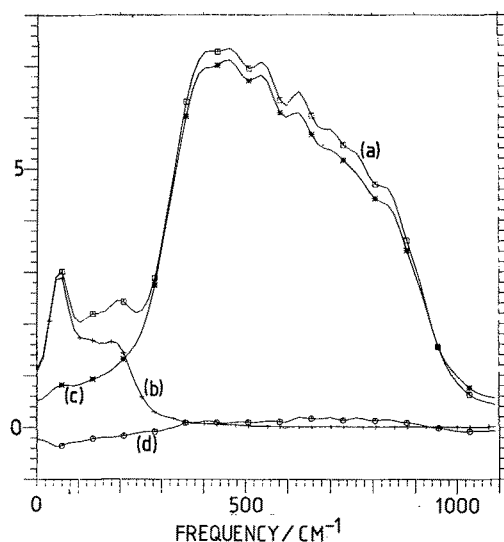


Figure 10. Inelastic neutron spectra (arbitrary units) for H_2O at 286 K. The outermost curve (a) shows the full spectrum and the other curves show its components: (b) centre of mass; (c) angular velocity; (d) cross term (see text).

where $\mathbf{R}_H$ is the displacement of the proton from the molecular centre of mass. It follows that it is possible to write $\tilde{Z}_H(\omega)$ as the sum of contributions from translations, rotations and rotation-translation coupling, that is as

$$\begin{aligned} \tilde{Z}_H(\omega) = \text{Re} \int_0^\infty \exp(i\omega t) \{ & \langle \mathbf{v}_{\text{CM}}(t) \cdot \mathbf{v}_{\text{CM}}(0) \rangle \\ & + \langle [\boldsymbol{\omega}(t) \times \mathbf{R}_H(t)] \cdot [\boldsymbol{\omega}(0) \times \mathbf{R}_H(0)] \rangle \\ & + 2 \langle \mathbf{v}_{\text{CM}}(t) \cdot \boldsymbol{\omega}(0) \times \mathbf{R}_H(0) \rangle \} dt. \end{aligned} \quad (15)$$

The three contributions are plotted separately in figure 10. The centre of mass contribution closely resembles $\tilde{Z}_o(\omega)$ (not shown) and has the expected peaks at around 60 and 190 cm^{-1} . Given the difficulties of the neutron scattering experiments, the positions of these peaks are in fair agreement with those observed in the experimental spectral density function (at 100 and 180 cm^{-1}) and assigned to centre of mass vibrations. It should be noted, moreover, that peaks assigned to centre of mass motion in infrared and Raman experiments [30] appear at 60 and 180 cm^{-1} .

From figure 10 it is clear that the broad band in $\tilde{Z}_H(\omega)$ at about 440 cm^{-1} arises from the librational motion of the molecules. In confirmation of this we find that for D_2O the band shifts to about 320 cm^{-1} , whereas the translational peaks move only slightly. As the earlier discussion of the infrared and Raman results would lead us to expect, the librational frequency predicted by the model is considerably too low. Experimentally the peak for H_2O is at 570 cm^{-1} , though

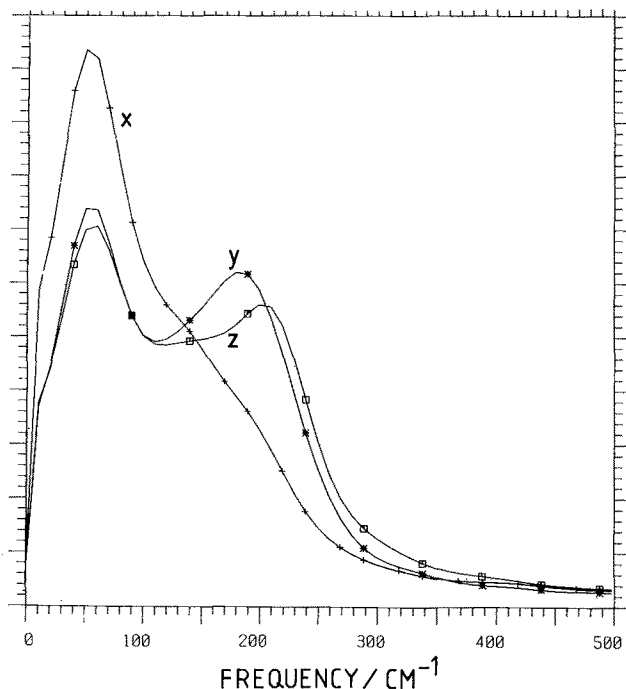


Figure 11. Fourier transforms of the autocorrelation functions of the components of the centre of mass velocity with respect to the molecular axes at the initial time (see figure 4).

this is still lower than the highest frequency infrared and Raman peak ($\simeq 700 \text{ cm}^{-1}$).

Figure 10 also shows that rotation-translation coupling makes only a small contribution to $\tilde{Z}_H(\omega)$. This does not imply that all manifestations of rotation-translation coupling are necessarily small. Indeed, we have already seen in § 4 that the observed large decrease in the self diffusion coefficient on deuteration may plausibly be ascribed to this effect.

In view of the good agreement between the frequencies of the peaks in $\tilde{Z}_H(\omega)$ assigned to centre of mass motion and those similarly assigned in the calculated infrared and Raman spectra, it is worthwhile to look more closely at the origin of these features, since they are clearly strongly characteristic of the potential model used, in addition to correlating well with experiment. To do so, we note that the term $\langle \mathbf{v}_{\text{CM}}(t) \cdot \mathbf{v}_{\text{CM}}(0) \rangle$ in (15) can be broken up into contributions arising from the three functions $Z^\alpha(t)$ defined by equations (3) and (4). The corresponding Fourier transforms $\tilde{Z}^\alpha(\omega)$ are plotted in figure 11. As anticipated in § 4, the two functions $\tilde{Z}^y(\omega)$ and $\tilde{Z}^z(\omega)$ prove to be very similar. Moreover, the three functions are characterized by only two well defined frequencies, the values of which correlate with the two centre of mass peaks in $\tilde{Z}_H(\omega)$. The resolution into components shows that the higher frequency is associated with in-plane oscillations of the molecule, whereas the lower one is linked to out-of-plane oscillations. In his analysis of the experimental spectroscopic data, Walrafen [30] assigns the high and low frequencies, respectively, to hydrogen bond stretching and bending.

7. DENSITY FLUCTUATIONS

The spectrum of density fluctuations in the liquid is described by the dynamical structure factor $S(k, \omega)$, defined as the Fourier transform of the density-density autocorrelation function by

$$S(k, \omega) = \int_0^\infty \exp(i\omega t) F(k, t) dt, \quad (16)$$

$$F(k, t) = \langle \sum_i \sum_j \exp\{-ik \cdot [\mathbf{R}_i(t) - \mathbf{R}_j(0)]\} \rangle, \quad (17)$$

where $\mathbf{R}_i, \mathbf{R}_j$ are the centre of mass coordinates of molecules i, j . For small k , $S(k, \omega)$ reduces to the familiar Rayleigh-Brillouin triplet. For large k the propagating sound mode responsible for the Brillouin lines is overdamped and $S(k, \omega)$ has a purely diffusive form. These two limits are common to all fluids, but the details of the transition from low k to high k behaviour are dependent on the form of the intermolecular potential. Defining a characteristic wave-number $k_\sigma = 2\pi/\sigma$, where σ is the mean nearest neighbour spacing, it is found that in argon-like liquids the sound mode disappears at about $k/k_\sigma = 0.2$, whereas in liquid metals it persists up to about $k/k_\sigma = 0.7$ [36].

In a computer simulation of the type undertaken here, use of periodic boundary conditions means that we are limited to the study of wavevectors of the form $\mathbf{k} = (2\pi/L)(l, m, n)$, where l, m, n are integers and L is the length of the primitive cube. For the calculations reported here, the lowest accessible wavenumber (at a molar volume of 18 cm^3) is $k_{\text{min}} = 0.183k_\sigma$, where $\sigma (= 2.84 \text{ \AA})$ is the position of the first peak in $g_{\text{OO}}(R)$. In figure 12 we show the results

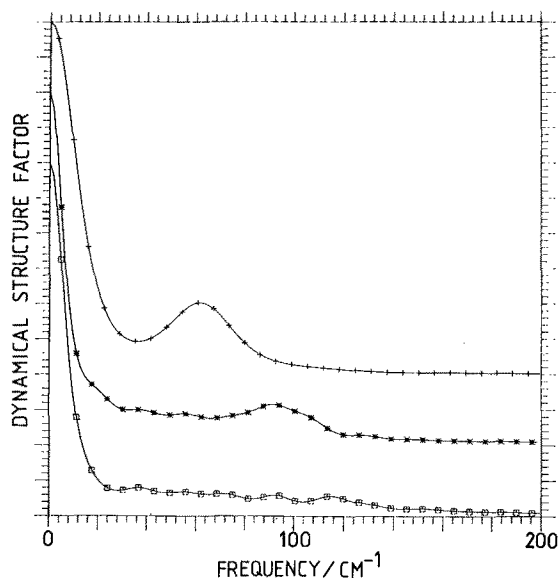


Figure 12. Dynamical structure factor for H_2O at 286 K for $(k/k_{\min})^2 = 1$ (top), 2 (middle), and 3 (bottom).

obtained for $S(k, \omega)$ for $(k/k_{\min})^2 = 1, 2$ and 3; these were calculated by numerically transforming the function $F(k, t)$ via a maximum entropy method [37]. All three spectra have a strong peak at $\omega = 0$. At the lowest wavenumber there is, in addition, a well developed second peak which broadens, shifts approximately linearly with k and eventually disappears as the wavenumber is increased. This is the remnant of the hydrodynamic Brillouin peak. Assuming linear dispersion, the frequency of the peak corresponds to a sound speed of 3040 m s^{-1} . This is high in comparison with the measured low- k value of 1490 m s^{-1} [38]. The discrepancy is not necessarily a defect of the MCY model, since there are a number of relaxation processes which could cause the liquid to appear stiffer at high wavenumbers. On the other hand, agreement with experiment is scarcely to be expected, since the simulation is one of a liquid under high pressure. The sound wave mode persists to larger k than in similar calculations for the ST2 potential [39], for which the Brillouin peak is only barely detectable at $k = 0.477 \text{ \AA}^{-1}$ ($k/k_{\sigma} = 0.22$). Spectra such as those shown in figure 12 are in principle measurable (for D_2O) by coherent inelastic scattering experiments. Given the differences apparent between the MCY and ST2 potentials, it is clear that such experiments could play a useful rôle in discriminating between models of the molecular interactions in water.

In their calculations for the ST2 potential, Raman and Stillinger [39] observed an enhanced intensity at high frequencies which they associated with a high frequency sound mode. Such an effect could arise from a strong coupling between the density fluctuations and correlations in molecular orientation extending over several molecular diameters. We have found no evidence of any such peak in our own work, which is consistent with the notion implicit in the earlier discussion that in certain respects the molecules in MCY water are less

strongly orientationally correlated than is the case for the ST2 potential. Broadly speaking, the density fluctuations for the MCY model are similar to those of a normal non-associated liquid.

8. DIELECTRIC PROPERTIES

8.1. Dielectric relaxation

In § 5, we mentioned some of the difficulties involved in the experimental observation of the long time decay of the single molecule orientational correlation functions. Dielectric relaxation experiments provide an example ; though they are sensitive to details of the reorientational process, it is difficult to relate the results directly to the properties of single molecules. Much attention is focused on extracting from the data an estimate for the Debye time τ_D , which is a quantity characterizing the decay of the collective orientation of the molecular dipole axes (in our case, the z axes). This is achieved by fitting the observed dispersion in ϵ to the function

$$\frac{\epsilon(\omega) - \epsilon_\infty}{\epsilon - \epsilon_\infty} = \frac{1}{1 - i\omega\tau_D}, \quad (18)$$

where ϵ_∞ is the permittivity in the far infrared region. For water τ_D is found experimentally to vary rapidly with temperature, suggestive of a strong dependence on the correlation of molecular orientations. At high frequencies the dispersion in ϵ gives rise to the librational bands in the infrared region which we have already discussed in § 5. As was emphasized there, however, it is desirable to check whether or not collective effects can significantly alter the character of the librational motion from that described by the single molecule reorientational functions.

Considerable care must be taken in relating the decay process observed in a dielectric relaxation experiment to properties calculated in a computer simulation. The main source of difficulty is the long range nature of the dipolar interaction. In particular [40], the total dipole moment of a small region within a sample does not relax as $\exp(-t/\tau_D)$, even in the rotational diffusion limit. Thus τ_D is not readily obtained from the rate of relaxation of the total dipole moment of the primitive cube, though this is the natural focus of attention in a computer simulation. For an isotropic fluid it can be shown [40] that the transverse part of the spatial Fourier transform of the microscopic dipole moment density, that is

$$\mathbf{M}_T(\mathbf{k}, t) = \sum_i [\boldsymbol{\mu}_i(t) \times \mathbf{k}] \exp[-i\mathbf{k} \cdot \mathbf{R}_i(t)], \quad (19)$$

where $\boldsymbol{\mu}_i$ is the dipole moment of molecule i , obeys a relation of the form

$$\frac{\epsilon(\omega) - \epsilon_\infty}{\epsilon - \epsilon_\infty} \propto \lim_{k \rightarrow 0} \left[1 + i\omega \int_0^\infty \exp(i\omega t) \Phi_T(k, t) dt \right], \quad (20)$$

where, for a classical system

$$\Phi_T(k, t) = \langle \mathbf{M}_T(\mathbf{k}, t) \cdot \mathbf{M}_T^*(\mathbf{k}, 0) \rangle / \langle \mathbf{M}_T(\mathbf{k}, 0) \cdot \mathbf{M}_T^*(\mathbf{k}, 0) \rangle. \quad (21)$$

The same analysis shows that none of the parameters characterizing the frequency dependence of $\Phi_T(k, \omega)$ depend upon long range dipole correlations. We may

therefore hope that the relaxation of $M_T(\mathbf{k}, t)$ for the infinite, periodic system used in the simulation will adequately represent the dielectric relaxation of an infinite, isotropic fluid of MCY molecules.

Two other problems should be noted. First, the use of periodic boundary conditions and the consequent lower limit on the accessible wavenumbers means that the limiting process in equation (20) can be taken only in an approximate way. Secondly the calculation of collective properties such as $M_T(\mathbf{k}, t)$ suffers in accuracy from the poor statistics involved and may also be sensitive to spurious orientational correlations induced by the periodic boundary conditions.

The autocorrelation function for the longitudinal component of the dipole moment density, that is

$$M_L(\mathbf{k}, t) = \sum_i [\boldsymbol{\mu}_i(t) \cdot \mathbf{k}] \exp[-i\mathbf{k} \cdot \mathbf{R}_i(t)] \quad (22)$$

can also be routinely calculated in a simulation. In the rotational diffusion limit it can be shown that $M_L(\mathbf{k}, t)$ behaves as

$$\lim_{k \rightarrow 0} \Phi_L(k, t) = \exp(-\epsilon t / \tau_D), \quad (23)$$

where

$$\Phi_L(k, t) = \langle M_L(\mathbf{k}, t) M_L^*(\mathbf{k}, 0) \rangle / \langle M_L(\mathbf{k}, 0) M_L^*(\mathbf{k}, 0) \rangle. \quad (24)$$

By contrast, in the same limit, the transverse component behaves as

$$\lim_{k \rightarrow 0} \Phi_T(k, t) = \exp(-t / \tau_D). \quad (25)$$

Thus for real water, for which $\epsilon \simeq 80$, $\Phi_L(k, t)$ will decay on a timescale about two orders of magnitude shorter than $\Phi_T(k, t)$. This means that short time effects may dominate the relaxation of the longitudinal component and yet have little influence on the decay of the transverse component. Lobo *et al.* [41] have used a model of rotational dynamics due to Nee and Zwanzig [42], which includes explicitly the effect of dipolar interactions, to show that collective propagating modes (dipolarons) could appear in strongly polar fluids. The presence of a dipolaron mode is expected to be seen most clearly in $\Phi_L(k, t)$.

To our knowledge the only simulations in which the correlation functions $\Phi_T(k, t)$ and $\Phi_L(k, t)$ have been computed are those of Pollock and Alder [43]. These authors have made a detailed study of the dielectric properties of the Stockmayer fluid. They observed a strong oscillation in $\Phi_L(k, t)$ which they ascribed to the dipolaron mode. In the Stockmayer model the only torques exerted are those due to dipole-dipole interactions. It is clearly of interest to determine whether or not dipolaron oscillations are observed in systems in which there are other contributions to the intermolecular torques, as is the case in any real dipolar fluid.

8.2. Behaviour of the transverse and longitudinal functions

Examples of the normalized autocorrelation functions describing the decay of the transverse and longitudinal moments (respectively, $\Phi_T(k, t)$ and $\Phi_L(k, t)$) are plotted in figure 13. The curves are for the smallest value of k accessible in the simulation ($k_{\min} = 0.183k_\sigma$) and are taken from the calculations for H_2O at 286 K, in which 754 wavevectors were included in the Ewald sum. However, the results shown are similar to those obtained in other calculations at around

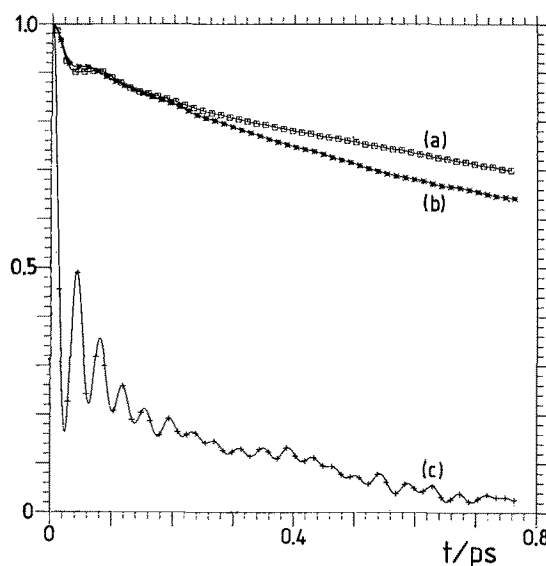


Figure 13. Dipole moment autocorrelation functions for H_2O at 286 K. From top to bottom : $C_1^z(t)$, $\Phi_T(k, t)$ and $\Phi_L(k, t)$.

room temperature and the only noticeable effect of increasing k , at least at small k , is to alter slightly the decay rate of $\Phi_T(k, t)$ at intermediate times.

The far infrared absorption coefficient is proportional to ω times the imaginary part of $\epsilon(\omega)$. Equation (20) shows that it is also proportional to ω^2 times the real part of the Fourier transform of $\Phi_T(k, t)$. The discussion of the high frequency absorption in § 5 was based on results obtained from equation (10). The assumption used there was that the collective function $\Phi_T(k, t)$ could to a good approximation be replaced by the single molecule function $C_1^z(t)$. The two functions are compared in figure 13; their similarity at intermediate times justifies the simplification used in § 5.

At large t , $\Phi_T(k, t)$ decays exponentially. The early onset of such behaviour is in accord with experimental results for water [32], which show that the relaxation of $\epsilon(\omega)$ is Debye-like over a very wide range of frequency. We shall denote the characteristic (exponential) decay time of $\Phi_T(k, t)$ by the symbol $\tau_T(k)$; note that it follows from equation (25) that

$$\lim_{k \rightarrow 0} \tau_T(k) = \tau_D. \quad (26)$$

For the function plotted in figure 12, $\tau_T(k)$ is approximately 3 ps. This result is much smaller than the experimental value of τ_D (12.6 ps at 10°C and 9.3 ps at 20°C [44]); the reasons for this discrepancy are discussed below.

The relationship between collective relaxation times, such as $\tau_T(k)$ and τ_D , and single molecule times, such as τ_1^z , has been investigated in [40]. It is shown there that the relationship may often be expressed in terms of an equilibrium orientational correlation parameter. For the case in hand, the relation of interest is

$$\tau_T(k) \simeq \tau_1^z g_T(k), \quad (27)$$

where the static correlation function $g_T(k)$ is given by

$$g_T(k) = 3 \langle |\mathbf{M}_T(\mathbf{k}, 0)|^2 \rangle / 2N\mu^2 \rho \\ = (3/2\mu^2) \langle \sum_j (\mu_1 \times \mathbf{k}) \cdot (\mu_j \times \mathbf{k}) \exp(-i\mathbf{k} \cdot \mathbf{R}_{1j}) \rangle \quad (28)$$

with $\mathbf{R}_{1j} = \mathbf{R}_1 - \mathbf{R}_j$. At low k equation (27) reduces to a modified Glarum-Powles formula

$$\tau_D \equiv \lim_{k \rightarrow 0} \tau_T(k) \simeq \tau_1^z \frac{3\epsilon}{2\epsilon + \epsilon_\infty} \quad (29)$$

where g_{Kirk} is the Kirkwood g -factor

$$\frac{(\epsilon - \epsilon_\infty)(2\epsilon + \epsilon_\infty)}{\epsilon} = \frac{4\pi N\tilde{\mu}^2}{Vk_B T} g_{\text{Kirk}} \quad (30)$$

and $\tilde{\mu}$ is the apparent dipole moment of the (polarizable) molecule.

From equation (29) it can be shown that the *calculated* value of τ_1^z given in table 2 is consistent with the *experimental* τ_D . The experimental values of g_{Kirk} , ϵ and ϵ_∞ , are respectively 2.55, 78.5 and 1.79 [44]; on inserting these in equation (29) we find that $\tau_D \simeq 3.8\tau_1^z = 12.2$ ps, in good agreement with the experimental results quoted above. Moreover, the calculated values of $\tau_T(k)$ are consistent with equation (27). For example, direct evaluation of $g_T(k)$ via equation (28) shows that $g_T(k_{\min}) \simeq 0.9$; the decay time $\tau_T(k_{\min})$ should therefore be slightly smaller than τ_1^z and this is confirmed by figure 13. What remains to be explained is why the calculated value of $\tau_T(k)$ for $k = k_{\min}$ is in such poor agreement with the experimental value for τ_D . There are two possibilities to be considered. First, the static dipolar correlations in the simulated liquid may not be representative of real water. Secondly, the low- k limit involved in (26) may be far from realized in a system of 125 molecules. For the particular molecular dynamics run under discussion, the static dielectric constant derived from (1) was $\epsilon \simeq 40$. If we introduce this value into (30), take $\epsilon_\infty = 1$ and set $\tilde{\mu}$ equal to the dipole moment for the MCY potential (which is consistent with the interpretation of the MCY model as an effective potential which implicitly takes account of polarization) we find that $g_{\text{Kirk}} \simeq 1.6$. This is much smaller than the experimental result. On the other hand, (27) and (29) imply that for the model fluid.

$$\lim_{k \rightarrow 0} g_T(k) = \frac{3\epsilon}{2\epsilon + 1} g_{\text{Kirk}}. \quad (31)$$

Hence the low- k limit of $g_T(k)$ should be approximately 2.4, which is considerably larger than the value obtained for $g_T(k)$ at the smallest accessible wavenumber. The conclusion we draw is that both the factors mentioned above play a significant role.

The question of system size effects is essentially a technical problem; it could be resolved if there were sufficient computing resources available to allow accurate calculations to be made for a much larger number of molecules in the primitive cell. The fact that the calculated values of ϵ and g_{Kirk} are considerably smaller than the experimental results is suggestive of an inadequacy in the MCY potential. However, the two factors are not independent. System size effects could well act so as to reduce the apparent dielectric constant of the

model, since there are likely to be important contributions to $g_T(k)$ from molecules separated by distances greater than half the length of the primitive cell.

As figure 13 shows, the longitudinal autocorrelation function $\Phi_L(k, t)$ for the smallest wavenumber is characterized by a rapid initial decay and a pronounced oscillation at intermediate times. The obvious question which arises is whether the oscillation is evidence of a dipolaron mode or of a librational motion of the water molecules which is linked to short range torques. It has been shown [40] that dipole moment correlation functions can develop oscillations under two distinct sets of conditions. If the molecules reorient slowly and the intermolecular torques are strong, oscillations may appear in all three of the functions $\Phi_T(k, t)$, $\Phi_L(k, t)$ and $C_1^z(t)$; it seems natural to label such behaviour as librational. At the other extreme, when the torques are weak and the decay time of the relevant function ($\Phi_T(k, t)$, $\Phi_L(k, t)$ or $C_1^z(t)$) is comparable with or shorter than the single particle free rotor time, an inertial oscillation may be observed; it is in this limit that the analysis of Lobo *et al.* [41] predicts the appearance of a dipolaron. The last condition cannot be satisfied for $C_1^z(t)$, since a molecule cannot reorient faster than in the free rotor limit, and it is unlikely to be satisfied both for $\Phi_T(k, t)$ and $\Phi_L(k, t)$, because the corresponding relaxation times differ by the factor ϵ . In the simulations of the Stockmayer fluid by Pollock and Alder [43] oscillations are seen only in $\Phi_L(k, t)$; the dipolaron description is appropriate here, because the torques are weak even when ϵ is large. In the present work oscillations of roughly the same frequency are seen in all three functions, as in figure 13; it is therefore more appropriate to describe the underlying motion as being librational in character. At the same time, it has to be recognized that the oscillations are most pronounced in $\Phi_L(k, t)$ precisely because the decay time of that function has been reduced onto the inertial timescale by the presence of the dipole correlation factor ϵ .

9. CONCLUSIONS

The results of the simulations can be expected to throw light on the properties of physical water only if the potential model used is a realistic one. It is therefore important to assess the extent to which the MCY potential is able to reproduce the known dynamical properties of water. From this point of view, the results we have obtained are generally satisfactory. Good overall agreement is found with the results of a variety of experiments, diffusion coefficient measurements, infrared and Raman spectroscopy, incoherent neutron scattering and dielectric relaxation. However, we have identified two areas in which improvement is needed. First, the calculated diffusion coefficients and reorientational correlation times vary too weakly with temperature. Secondly, the highest librational frequency seen experimentally is absent in the simulations. Our analysis shows not only that the latter feature of the calculations implies that the intermolecular torques are in certain respects too weak, but that it is correlated with the fact that the reorientation of the molecule about different axes has too isotropic a character. Thirdly, there is some indication that the static dipolar correlations are not fully representative of real water, since the values obtained for the dielectric constant and Kirkwood g -factor are in both cases too low. However, the study of dielectric properties is complicated by the use of periodic

boundary conditions and firm conclusions must await the results of calculations for a much larger system.

The conclusion which we draw about the properties of real water may conveniently be summarized under two headings.

Translational motion. The pronounced effect of deuteration on the magnitude of the diffusion coefficient appears to be largely explicable in classical terms. The peaks at about 60 and 180 cm^{-1} in the experimental infrared and Raman spectra arise from translational motions which we assign, respectively, to vibrations of the centre of mass perpendicular and parallel to the molecular plane. Where there is conflict between the results of optical and neutron scattering experiments our results support the former.

Molecular reorientation. A large isotope effect on the reorientational correlation times is predicted even for a purely classical system, though difficulties both in the molecular dynamics calculations and in the interpretation of the experimental data mean that the practical significance of this result is unclear. The main features of the high frequency regions in both the infrared and Raman spectra can be related in a straightforward way to the motion of single molecules. A satisfactory fit to the experimental spectroscopic data, inclusive of temperature effects, is obtained with a model which makes use of the gas phase polarizabilities and dipole moments and takes no account of hydrogen bond stretching or similar distortions. It appears that the single molecule dipole reorientation time can be obtained from the dielectric relaxation time *via* a relation involving only static correlations, though the evidence for this is somewhat indirect. The autocorrelation function for the longitudinal component of the total dipole moment density can be expected to show a strong oscillation which we would characterize as librational rather than dipolaron-like. Such an oscillation should be observable by infrared reflectance spectroscopy.

Finally, the calculations we have made of the spectrum of density fluctuations in the simulated liquid suggest that the results of coherent inelastic scattering experiments on water should resemble those obtained for simpler liquids. However, this conclusion is model dependent and our results do not agree well with those previously obtained for the ST2 potential.

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